Code and Data for "Structured Factor Regression with High-Dimensional Multimodal Data"

Overview

This repository contains the R code and data required to reproduce the simulation studies and real data analysis presented in the paper *Structured Factor Regression with High-Dimensional Multimodal Data*.

It includes the implementation of the proposed method, alternative methods for comparison, simulation and real data analysis scripts, as well as the corresponding datasets and precomputed results.

Repository Structure

The repository is organized as follows:

```
code_and_data/
  InferMultiModal/                      # R package implementing the proposed method
     R/
        betatest.R
        estimation.R
        gammatest.R
        predict.estimation.R
        rcv.R
     man/                                # Package help documentation
     DESCRIPTION
     NAMESPACE
     InferMultiModal.Rproj

  InferMultiModal.pdf                   # Package help manual

  alternative/                          # Implementations of alternative methods
     beta_fan.R
     beta_li.R
     joint_func.R

  Data/                                 # Data used in the real-data analysis
     RealData/
        RawData/                       # Raw TCGA data
           luad_tcga/
           lusc_tcga/
        ProcessedData/                 # Processed real-data dataset
            lung_data.RData
            processed_procedure.R

  simulation/                           # Simulation studies
     Replication_code/                 # Scripts for simulation experiments
        MainSetting/
           MainSetting_estimation.R
           MainSetting_betatest.R
           MainSetting_gammatest.R
           R_sensitivity/
               R_sensitivity_estimation.R
               R_sensitivity_betatest.R
               R_sensitivity_gammatest.R
        WeakSetting/
```

```
            WeakSetting_estimation.R
            WeakSetting_betatest.R
            WeakSetting_gammatest.R
        OnlyCommon/
            onlycommon_estimation.R
            onlycommon_betatest.R
        NoCommon/
            nocommon_estimation.R
            nocommon_betatest.R
        LowerNoiseSetting/
             LowerNoiseSetting_estimation.R
             LowerNoiseSetting_betatest.R
             LowerNoiseSetting_gammatest.R

    Results/                          # Precomputed simulation results
        betacsv/                      # Residual-effect testing results
        estimationcsv/                # Estimation and prediction results
        gammacsv/                     # Factor-effect testing results

    Fig_Tab_Code/                     # Scripts for generating figures and tables
         Tab1.R
         TabS1.R
         TabS2.R
         TabS3.R
         TabS4.R
         TabS5.R
         Fig1 and FigS1.R
         Fig2 and FigS2.R
         Fig3 and FigS3.R
         Fig4 and FigS4.R
         FigS5-S6,S8-S9.R
         FigS7.R
         FigS10 and FigS11.R
         FigS12, FigS13 and FigS14.R
         FigS15, FigS16 and FigS17.R
         FigS18 and FigS19.R
         FigS20.R
         FigS21.R
         FigS22 and FigS23.R
         FigS24.R
         FigS25.R
         FigS26.R
         FigS27.R
         FigS28.R
         FigS29.R

realdata/                             # Real-data analyses
    Full_Sample_Analysis/             # Full-sample estimation and inference
        Whole_Sample.R
        Fig5.R
        TabS6.R
    Resampling_Analysis/              # Resampling-based analysis
        Prediction_Stability.R
```

```
        averageError_and_jaccard.R
    Results/                          # Precomputed real-data results

  InferMultiModal_code_and_data.Rproj  # RStudio project file
  ReadMe.pdf                          # PDF version of this README
```

1. R Package: `InferMultiModal`

The `InferMultiModal/` directory contains the R package implementing the proposed method. The main functions include:

- `estimation()`: Estimates the parameters of the proposed factor regression model.
- `betatest()`: Tests the effects of idiosyncratic residuals.
- `gammatest()`: Tests the effects of common and modality-specific factors.
- `predict.estimation()`: Generates predictions for new observations based on a fitted model.

The procedure first estimates the factor components from the multimodal data. These estimated components are then used in the subsequent model estimation and inference steps. The factor estimation procedure is implemented separately, and its results are supplied to `estimation()` through the `res_fac` argument.

Installation

The package can be installed:

```r
devtools::install(here::here("InferMultiModal"))
library(InferMultiModal)
```

For detailed function descriptions, argument specifications, and usage examples, see the package help documentation in `InferMultiModal.pdf`.

2. Alternative Methods

The `alternative/` directory contains the implementations of alternative methods used for comparison with the proposed approach.

The main files include:

- `beta_li.R`: Implementation of the integrative factor regression method proposed by Li and Li (2022).
- `beta_fan.R`: An adapted implementation of the method proposed by Fan et al. (2024), based on their publicly available code.
- `joint_func.R`: Functions for joint estimation.

3. Simulation Studies

The `simulation/` directory contains the scripts for conducting simulation experiments, the corresponding precomputed results, and the scripts for generating figures and tables reported in the main manuscript and supplementary materials.

3.1 Simulation Settings

Five simulation settings are considered:

- **Main Setting:** The primary simulation scenario.
- **Weak Setting:** A scenario with weaker factors.
- **Only-Common-Factor Setting:** A scenario in which only the common factors are present.
- **No-Common-Factor Setting:** A scenario without the common factors.
- **Lower-Noise Setting:** A scenario with a lower noise level (higher-signal-to-noise).

The corresponding scripts are organized into separate subdirectories under `simulation/Replication_code/`.

3.2 Replication Code

The `simulation/Replication_code/` directory contains the R scripts for conducting simulation experiments under different settings. For each setting, the scripts are organized according to the following tasks:

- `*_estimation.R`: Performs parameter estimation and prediction.
- `*_betatest.R`: Conducts hypothesis testing for the effects of idiosyncratic residuals.
- `*_gammatest.R`: Conducts hypothesis testing for factor effects.

The scripts are organized into the following subdirectories:

Main Setting (`MainSetting/`) - `MainSetting_estimation.R` - `MainSetting_betatest.R` - `MainSetting_gammatest.R`

Weak Setting (`WeakSetting/`) - `WeakSetting_estimation.R` - `WeakSetting_betatest.R` - `WeakSetting_gammatest.R`

Only-Common-Factor Setting (`OnlyCommon/`) - `onlycommon_estimation.R` - `onlycommon_betatest.R`

No-Common-Factor Setting (`NoCommon/`) - `nocommon_estimation.R` - `nocommon_betatest.R`

Lower-Noise Setting (`LowerNoiseSetting/`) - `LowerNoiseSetting_estimation.R` - `LowerNoiseSetting_betatest.R` - `LowerNoiseSetting_gammatest.R`

3.3 Sensitivity Analysis

Sensitivity analyses are conducted under the main simulation setting to investigate the impact of latent dimension misspecification on the performance of the proposed method.

The corresponding scripts are located in `simulation/Replication_code/MainSetting/R_sensitivity/`:

- `R_sensitivity_estimation.R`: Performs parameter estimation and prediction under different choices of latent dimensions.
- `R_sensitivity_betatest.R`: Evaluates the hypothesis testing procedure for idiosyncratic residual effects.
- `R_sensitivity_gammatest.R`: Evaluates the hypothesis testing procedure for factor effects.

3.4 Precomputed Simulation Results

The `simulation/Results/` directory contains precomputed simulation results, including numerical results stored in CSV files and figures stored in PNG files.

- `betacsv/`: Results associated with hypothesis testing for idiosyncratic residual effects.
- `estimationcsv/`: Results associated with parameter estimation and prediction.
- `gammacsv/`: Results associated with hypothesis testing for factor effects.

These precomputed results are used as inputs by the figure- and table-generation scripts.

3.5 Figures and Tables

The `simulation/Fig_Tab_Code/` directory contains the R scripts for generating the figures and tables reported in the main manuscript and supplementary materials. The scripts are named according to their corresponding figure and table numbers, and some scripts generate multiple outputs. The complete mapping between scripts and their outputs is provided below.

Script	Output	Description
`Tab1.R`	Table 1	Empirical rejection rates (size and power) of the proposed and FARM tests for β under the main setting.

Script	Output	Description
`Fig1 and FigS1.R`	Figures 1 and S1	Comparison of estimation errors under Gaussian (Figure 1) and uniform (Figure S1) noise.
`Fig2 and FigS2.R`	Figures 2 and S2	Comparison of prediction errors under Gaussian (Figure 2) and uniform (Figure S2) noise.
`Fig3 and FigS3.R`	Figures 3 and S3	Empirical rejection rates (size and power) of the proposed and IFRM tests for β_k's under Gaussian (Figure 3) and uniform (Figure S3) noise.
`Fig4 and FigS4.R`	Figures 4 and S4	Empirical rejection rates (size and power) of the proposed test for γ_c and of the proposed and IFRM tests for $\gamma_{s,k}$'s under Gaussian (Figure 4) and uniform (Figure S4) noise.
`FigS5-S6,S8-S9.R`	Figures S5, S6, S8, and S9	Comparison of estimation errors (Figures S5 and S8) and prediction errors (Figures S6 and S9) under Gaussian noise in the no-common-factor and only-common-factor settings, respectively.
`FigS7.R`	Figure S7	Empirical rejection rates (size and power) of the proposed and IFRM tests for β_k's under the no-common-factor setting.
`FigS10 and FigS11.R`	Figures S10 and S11	Comparison of estimation errors (Figure S10) and prediction errors (Figure S11) among the proposed method, joint estimation, and oracle procedures under the main setting.
`FigS12, FigS13 and FigS14.R`	Figures S12, S13, and S14	Empirical rejection rates (size and power) for β_k's, comparing the proposed method with joint estimation (Figure S12), the oracle-factor-number-based method (Figure S13), and the oracle method (Figure S14) under the main setting.
`FigS15, FigS16 and FigS17.R`	Figures S15, S16, and S17	Empirical rejection rates (size and power) for γ_c and $\gamma_{s,k}$'s, comparing the proposed method with joint estimation (Figure S15), the oracle-factor-number-based method (Figure S16), and the oracle method (Figure S17) under the main setting.

Script	Output	Description
`FigS18 and FigS19.R`	Figures S18 and S19	Comparison of estimation errors (Figure S18) and prediction errors (Figure S19) under the weaker setting.
`FigS20.R`	Figure S20	Empirical rejection rates (size and power) of the proposed and IFRM tests for β_k's under the weaker setting.
`FigS21.R`	Figure S21	Empirical rejection rates (size and power) of the proposed test for γ_c and of the proposed and IFRM tests for $\gamma_{s,k}$'s under the weaker setting.
`FigS22 and FigS23.R`	Figures S22 and S23	Comparison of estimation errors (Figure S22) and prediction errors (Figure S23) under the lower-noise setting.
`FigS24.R`	Figure S24	Empirical rejection rates (size and power) of the proposed test for β_k's under the lower-noise setting.
`FigS25.R`	Figure S25	Empirical rejection rates (size and power) of the proposed test for γ_c and of the proposed and IFRM tests for $\gamma_{s,k}$'s under the lower-noise setting.
`FigS26.R`	Figure S26	Comparison of estimation errors under different specifications of the factor dimensions r.
`FigS27.R`	Figure S27	Comparison of prediction errors under different specifications of the factor dimensions r.
`FigS28.R`	Figure S28	Empirical rejection rates (size and power) of the proposed test and its variants with prespecified r for β_k's under Gaussian noise and Gaussian factor loadings.
`FigS29.R`	Figure S29	Empirical rejection rates (size and power) of the proposed test and its variants with prespecified r for γ_c and $\gamma_{s,k}$'s under Gaussian noise and Gaussian factor loadings.
`TabS1.R`	Table S1	Empirical rejection rates (size and power) of the proposed and FARM tests for β under the only-common-factor setting.
`TabS2.R`	Table S2	Empirical rejection rates (size and power) of the proposed and alternative tests for β under the main setting, comparing joint estimation and oracle procedures.

Script	Output	Description
`TabS3.R`	Table S3	Empirical rejection rates (size and power) of the proposed and alternative tests for β under the weaker setting.
`TabS4.R`	Table S4	Empirical rejection rates (size and power) of the proposed and alternative tests for β under the lower-noise setting.
`TabS5.R`	Table S5	Empirical rejection rates (size and power) of the proposed test and its variants with prespecified r for β under Gaussian noise and Gaussian factor loadings.

4. Real Data Analysis

The `realdata/` directory contains the scripts for the real data analyses, including full-sample analyses and resampling-based analyses. The corresponding datasets are stored in `Data/RealData/`, with raw and processed data organized separately.

The real data analyses are based on two TCGA lung cancer cohorts: lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC). Four types of molecular data are considered: mRNA expression, DNA methylation, copy number alteration (CNA), and protein expression. The clinical outcome is the prebronchodilator forced expiratory volume in one second (FEV1).

4.1 Full-Sample Analysis

The `realdata/Full_Sample_Analysis/` directory contains scripts for estimation and inference based on the full sample using the proposed method.

- `Whole_Sample.R`: Performs model estimation and hypothesis testing using the processed real-data dataset and the proposed method.
- `Fig5.R`: Generates Figure 5 in the main manuscript using the precomputed results stored in `realdata/Results/`.
- `TabS6.R`: Generates Table S6 in the supplementary materails using the precomputed results stored in `realdata/Results/`.

4.2 Resampling Analysis

The `realdata/Resampling_Analysis/` directory contains scripts for evaluating prediction performance and the stability of the analysis results through resampling.

- `Prediction_Stability.R`: Performs the resampling analysis using repeated random train-test splits with a training-to-test sample size ratio of 20:1 to evaluate prediction performance and the stability of the results. This script requires rerunning the resampling procedure and may be computationally intensive.
- `averageError_and_jaccard.R`: Calculates the average prediction error and Jaccard indices based on the precomputed resampling results stored in `realdata/Results/`. It produces numerical results reported in the manuscript and does not require rerunning the resampling procedure.

4.3 Precomputed Real-Data Results

The `realdata/Results/` directory stores the precomputed results from the full-sample and resampling analyses. These results are used directly by `Fig5.R`, `TabS6.R`, and `averageError_and_jaccard.R` to generate

the corresponding figures, tables, and numerical summaries without rerunning the computationally intensive analyses.

5. Data Availability

5.1 Data Sources

The original LUAD and LUSC data are publicly available through cBioPortal. The corresponding TCGA studies are LUAD and LUSC.

Due to file size limitations, the downloaded raw data files are not included in this repository. To reproduce the preprocessing procedure, users need to download the raw data for both studies from cBioPortal and extract the files into the corresponding directories under `Data/RealData/RawData/`, organized into separate `luad_tcga/` and `lusc_tcga/` folders.

5.2 Data Preprocessing

The preprocessing script `processed_procedure.R` is provided in `Data/RealData/ProcessedData/`. After downloading and extracting the raw data into the directories specified in Section 5.1, users can run this script to reproduce the preprocessing steps, including extracting the clinical outcome, processing the molecular data, matching samples across data types, and constructing the final analysis dataset.

The resulting processed dataset, `lung_data.RData`, used in the analyses is stored in `Data/RealData/ProcessedData/`.

5.3 Processed Data Structure

The processed real-data dataset is stored as a data frame named `Lung_df` in an RData file, combining the LUAD and LUSC cohorts. It contains 241 observations and 1,095 variables, including one clinical outcome and 1,094 molecular features from four data modalities.

The variables are organized as follows:

Variable type (Number)	Description
`FEV1_PERCENT_REF_PREBRONCHOLIATOR` (1)	prebronchodilator forced expiratory volume in one second (FEV1, clinical outcome)
mRNA expression (300)	Selected mRNA expression features
DNA methylation (300)	Selected DNA methylation features
Copy number alteration (CNA) (300)	Selected CNA features
Protein expression (194)	Selected protein expression features
Total (1,095)	

Each row corresponds to a TCGA sample, with sample identifiers provided as row names. The processed dataset serves as the input for the real-data analyses.

6. Reproducibility

All file paths in the scripts are specified relative to the project root using the `here` package. To ensure that the paths are correctly resolved, open the `code_and_data` directory as an RStudio Project before running the scripts. No manual working-directory setting using `setwd()` is required.

To reproduce the analyses, please follow the steps below:

1. **Open the project.** Open `InferMultiModal_code_and_data.Rproj` in RStudio.
2. **Install the required R packages.** Install the R packages required by the corresponding scripts.
3. **Install the proposed-method package.** Install the `InferMultiModal` package from the local repository directory and load it before running the analysis scripts.

4. **Reproduce the simulation experiments.** Run the scripts in `simulation/Replication_code/` under the corresponding simulation settings. These scripts perform parameter estimation, prediction, and hypothesis testing.
5. **Generate simulation figures and tables.** Run the scripts in `simulation/Fig_Tab_Code/` to generate the corresponding figures and tables using the precomputed results in `simulation/Results/`. The simulation experiments do not need to be rerun to reproduce these outputs.
6. **Prepare the real-data dataset.** If the processed dataset is not available, obtain the required raw data from cBioPortal and run `Data/RealData/ProcessedData/processed_procedure.R` to construct the analysis dataset.
7. **Reproduce the full-sample real-data analysis.** Run `realdata/Full_Sample_Analysis/Whole_Sample.R` to perform estimation and hypothesis testing using the proposed method.
8. **Reproduce the resampling analysis.** Run `realdata/Resampling_Analysis/Prediction_Stability.R` to conduct the computationally intensive resampling procedure. Alternatively, use the provided precomputed results to reproduce the numerical summaries without rerunning the resampling analysis.
9. **Generate real-data figures, tables, and numerical summaries.** Run the following scripts:
    - `realdata/Full_Sample_Analysis/Fig5.R`
    - `realdata/Full_Sample_Analysis/TabS6.R`
    - `realdata/Resampling_Analysis/averageError_and_jaccard.R`

These scripts use the precomputed results stored in `realdata/Results/` and do not require rerunning the corresponding analyses.

References

- Li, Q., & Li, L. (2022). Integrative factor regression and its inference for multimodal data. *Journal of the American Statistical Association*, 117(540), 2207–2221.
- Fan, J., Lou, Z., & Yu, M. (2024). Are latent factor regression and sparse regression adequate? *Journal of the American Statistical Association*, 119(546), 1076–1088.